

Research Roadmap and Implementation Plan

Research Topic

Development of an Enhanced Similarity Degree Calculation Procedure for Multiclass Imbalanced Datasets

1. Introduction

This research focuses on improving the Similarity Degree Algorithm, a core concept used in the SOUP technique for multiclass imbalanced data. The goal is to design a more robust, interpretable, and scalable similarity degree computation method that enhances resampling accuracy and classifier performance.

2. Research Objectives

- Develop an enhanced similarity degree algorithm for multiclass imbalanced datasets.
- Incorporate data difficulty factors and feature importance into similarity computation.
- Validate the new algorithm against established resampling methods such as SMOTE and SOUP.
- Evaluate performance using metrics such as F-Score, AUC, and G-Mean.

3. Research Roadmap

Phase 1: Literature Review

- Review existing techniques for multiclass imbalance handling (e.g., SMOTE, SOUP, MIRT).
- Identify limitations in current similarity degree approaches.
- Explore mathematical similarity measures such as Euclidean, Cosine, Mahalanobis, and Kernel-based similarities.

Phase 2: Algorithm Design

- Preprocess data (handle missing values, normalize features).
- Compute feature weights using feature importance methods (Random Forest, SHAP values).

- Cluster class instances to identify subgroups within each class.
- Use Mahalanobis or kernel-based metrics to measure inter-class distance.
- Convert distances to similarity scores using exponential decay functions.
- Introduce regularization to prevent extreme similarity values.

Phase 3: Implementation

- Implement the enhanced algorithm in Python using libraries such as NumPy, pandas, and scikit-learn.
- Generate similarity matrices and visualize with heatmaps and dendrograms.
- Integrate the algorithm with resampling frameworks to improve data balancing.

Phase 4: Evaluation

- Compare the enhanced similarity degree approach with existing methods on benchmark datasets.
- Analyze metrics such as F-Score, AUC, G-Mean, and classifier interpretability.
- Test scalability on high-dimensional datasets.

Phase 5: Documentation and Publication

- Prepare detailed documentation of methodology, results, and analysis.
- Publish results in machine learning and data mining journals.
- Propose extensions for dynamic similarity and cross-domain analysis.

4. Proposed Enhanced Similarity Degree Algorithm

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- Step 1: Data Preprocessing -- Normalize and encode features, handling missing data.
 - Step 2: Feature Weighting -- Compute feature importance and apply weighted similarity.
 - Step 3: Cluster Analysis -- Identify subgroups within classes using hierarchical clustering.
 - Step 4: Distance Metric -- Use weighted Mahalanobis or kernel-based metrics for distance.

- Step 5: Similarity Computation -- Convert distances to similarities using exponential decay.
- Step 6: Regularization -- Apply a baseline to smooth extreme values.
- Step 7: Normalization -- Normalize similarity scores between 0 and 1.
- Step 8: Visualization -- Generate heatmaps and dendrograms for class relationships.
- Step 9: Validation -- Compare results with baseline similarity methods and evaluate classifier performance.

5. Expected Outcomes

- A robust algorithm for similarity degree computation applicable to multiclass imbalanced data.
- Improved accuracy and fairness in resampling-based classification.
- Enhanced interpretability of class relationships through visualization tools.
- Published research contribution in computational intelligence and data science.